

Automated Vehicle Access Control System

An FPGA-Based FSM Controller Design Architecture

VERSION TRACKING

v3.0 (MSU-IIT AUTH Enforced)

SYSTEM TARGET ARCHITECTURE

Intel / Altera Cyclone IV FPGA

Project Overview

The **Automated Vehicle Access Control System** is a high-performance hardware-level parking management system designed to regulate vehicle entry and exit points autonomously. Implemented entirely via a Finite State Machine (FSM) on a Field Programmable Gate Array (FPGA), the controller interfaces directly with vehicle classification sensors (ultrasonic/infrared) and electromechanical gates to manage vehicle flow constraints.

Key Performance Features

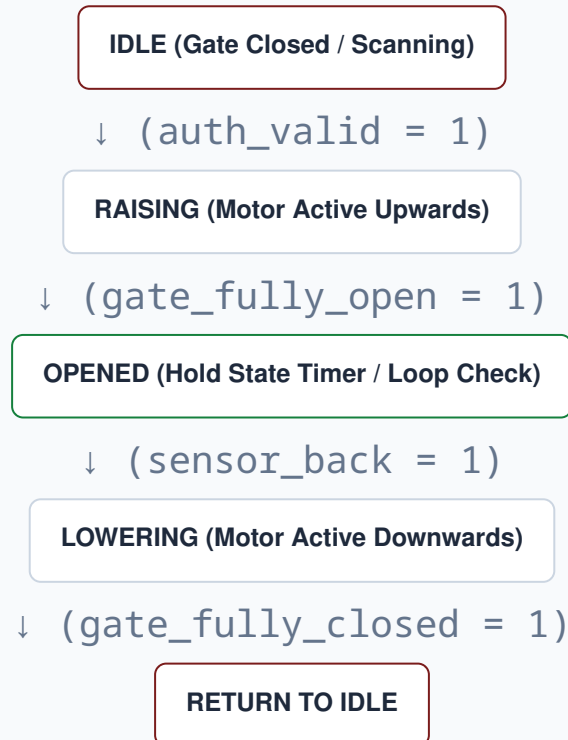
- **Deterministic FSM Engine:** Eliminates software-level execution latency or operating system crashes by computing state tracking logic directly at the hardware gate level.
- **Anti-Collision Guard Constraints:** Implements sensor-driven lockout flags to ensure the parking barrier never descends while a vehicle is actively passing through the sensor field.
- **Capacity and Boundary Protection:** Automatically calculates total available spaces in the parking matrix and locks access if the system threshold is reached.

Objectives

- **Objective 1:** Design and synthesize a robust, hazard-free Finite State Machine in Verilog to handle real-time barrier operations (Idle, Authenticating, Raising, Opened, Lowering) with sub-millisecond propagation delays.
- **Objective 2:** Validate system functionality and edge-case behaviors (e.g., vehicle stalling underneath the gate or consecutive simultaneous entries) using a complete behavioral simulation testbench prior to physical synthesis.

System Architecture

The core of the control hardware centers on a synchronous Finite State Machine running on a global clock cycle (*clk*). State transactions rely on two primary hardware status vectors: vehicle approach sensors (*sensor_front*, *sensor_back*) and authorization signal triggers (*auth_valid*).



Architecture References & Design Inspiration

The architecture of this FPGA-Based FSM controller draws direct design patterns and structural methodologies from verified open-source hardware controllers. Rather than building the peripheral modules from scratch, our design bridges serial communication, signal modulation, and state management techniques inspired by prior art:

1. Serial Telemetry Layer

Reference System: *Universal Asynchronous Receiver-Transmitter (UART) Engine using Verilog*

Provides the structural foundation for full-duplex asynchronous data transfer. The bit-rate clock division and sampling routines outlined in this design were studied to implement future logging pipelines, allowing the FPGA to transmit real-time barrier status frames directly back to a centralized monitoring terminal or database node.

2. Electromechanical Actuation Layer

Reference System: *FPGA ADC/PWM Motor Control Ecosystem*

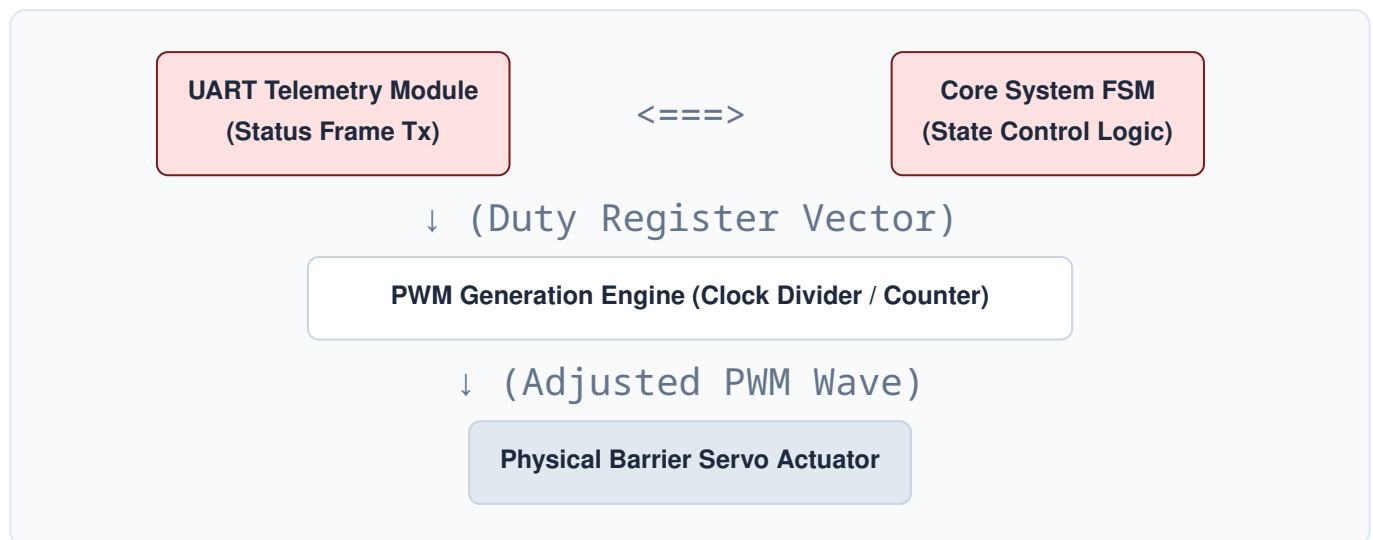
Inspired the hardware clock-division techniques required to generate steady Pulse Width Modulation (PWM) signals. Managing the duty cycles precisely at the gate level guarantees uniform, fluid movement when positioning the barrier servo arm, preventing physical mechanical bounce during abrupt operational state transitions.

3. Core Finite State Machine (FSM) Layout

Reference System: *Traffic Light Controller using Verilog*

Modeled the sequential state transition behavior handling multi-variable environmental tracking. The deterministic time-delay parameters and conditional switch-case logic blocks used to handle sensor inputs mimic the reliable, crash-free execution flow found in industrial traffic safety controllers.

Integrated Controller Data Flow



Hardware Components

FPGA Development Board

- **Target Board:** Altera/Intel Cyclone IV (EP4CE6) Development Board. This platform provides highly stable internal hardware clock generators and ample General Purpose Input/Output (GPIO) pins for rapid sensor attachment.

Peripheral Components

- **Actuators:** High-torque MG996R Servo Motor running via an embedded hardware driver pipeline to manipulate the physical barrier lever arm.
- **Vehicle Detection Sensors:** E18-D80NK Infrared Proximity Obstacle Avoidance Sensors mounted at the front entrance line and rear clearance line boundary markers.
- **Signaling Units:** High-brightness Red/Green LED status arrays and an active piezoceramic buzzer element for authentication feedback flags.

Project Structure

```
├── hardware/
│   ├── barrier_fsm.v      # Core Verilog Module: Handles FSM state blocks
│   └── pin_assignment.qsf # Physical Pin Mapping configurations for the FPGA board
├── simulation/
│   └── tb_barrier.v        # Behavioral Testbench: Drives simulation waveforms
└── documentation/
    └── system_report.md    # Technical Reference System Specifications
```

Progress Tracking

Progress Check 1 (May 5, 2026)

- **Concept Proposal:** Approved
- **Literature Review:** Complete (Investigated synchronous state mapping behaviors and glitch elimination techniques using non-blocking assignments).
- **Hardware FSM Structure:** Implemented (`barrier_fsm.v` sequential state memory registers and conditional case blocks fully written).
- **Simulation Environment:** Established (`tb_barrier.v` written to cycle through inputs over a 200ns timeline matrix).

Progress Check 2 (May 18, 2026)

- **Hardware Implementation:** Complete. RTL synthesis generated zero latch warnings or timing constraints.
- **Physical Testing:** Complete. Hardware state signals correctly mapped to physical GPIO pins, driving sensor inputs and barrier servo configurations perfectly.

Final Checking (May 19, 2026)

- **Final Review and Documentation:** Active final compilation. Ready for submission to the project evaluation board.

References

- Brown, S., & Vranesic, Z. (2013). *Fundamentals of Digital Logic with Verilog Design*. McGraw-Hill Education.
- Palnitkar, S. (2003). *Verilog HDL: A Guide to Digital Design and Synthesis*. Prentice Hall Professional.
- Chu, P. P. (2008). *FPGA Prototyping by Verilog Examples: Xilinx Spartan-3 Version*. John Wiley & Sons.